

Research Proposal: An Outline for PhD

Roshan Raj

May, 2024

Tentative Title

On mathematical aspects of Quantum Field Theory

Context and Purpose of the Project

Historical attempts by notable physicists like Heisenberg, Dirac, and von Neumann laid the groundwork for quantum mechanics. However, Quantum field theory (QFT) still suffers from unresolved issues such as non-renormalizability and the limitations of perturbative methods. QFT is a foundational theory in physics that describes how particles interact and form the basis for the Standard Model. Despite its empirical success, QFT lacks a rigorous mathematical foundation. This research aims to identify a set of axioms that can provide a consistent and comprehensive mathematical framework for QFT, addressing the current theoretical challenges and inconsistencies.

Underlying Basis of the Project

As Michio Masujima describes in his book [1], “There is a Hamiltonian formalism and a Lagrangian formalism for quantum mechanics, just as for analytical mechanics. The wave mechanics of Schrodinger and the matrix mechanics of Heisenberg belong to the Hamiltonian formalism, while the path integral quantization of Feynman, which originates from Dirac, belongs to the Lagrangian formalism.”

However, a few years after Path integral formalism, Haag’s theorem pointed out that QFT is weak at its core and requires better mathematical rigour [2], especially when one uses perturbation theories in addition to its other drawbacks such as non-renormalizability and limited validity of Feynmann’s diagrams appears. Therefore, It saw problems in perturbative quantum field theories, making QFT use heuristically as a guiding principle. However, several works were done to make the Standard Model, which brought about a renaissance in QFT and subdued most problems of perturbative quantum field theories except for a lack of mathematical rigour.

The project is motivated by the need for a robust mathematical framework that can unify and rigorously support the principles of QFT. The open area of research, “axiomatic approach to QFT,” aims for mathematical rigour, formally put, “What is the particular set of Axioms which can completely express the Quantum field theory mathematically?” [3]. The axiomatic approach to QFT has pursued trial & error methodology to analyze and resolve the problem by considering different sets of axioms at different points of the era [4]. In such QFT, the emphasis has shifted from quantum fields to local observables (operators). Wightman Axioms, which introduced conditions for constructing QFT, were replaced by the Haag-Kastler Axioms, which focused on the algebraic structure of local observables; then, Osterwalder-Schrader Reconstruction connected Euclidean field theory with Minkowskian QFT. Recent research has focused on algebraic QFT [5], topological aspects of QFT, and FQFT (Functorial QFT (1990s-present), which focuses on states rather than observables [6].

Each approach has contributed to our understanding, but a universally accepted set of axioms remains elusive. By exploring axiomatic approaches, this research seeks to build on these foundational efforts and address the long-standing dilemmas in QFT.

Methodology

1. Analyze existing axiomatic approaches, highlighting their similarities and differences to pinpoint the reasons behind their limitations.

2. Develop a mathematically consistent or infinitesimal free formalism of classical field theory, which can potentially reflect certain general features that an axiomatic field theory should need to carry, though classically. Then, pluck out a set of axioms embedded into classical theory that can necessarily help guess a similar set of axioms for QFT.
3. If needed, opting for a shift in the definition, physical understanding and representation of the observable based on the clear distinctiveness of quantum fields theory and quantum mechanics.
4. Execute the above steps to construct the relativistically & quantum physically consistent set of axioms for quantum field theory.

Expected Results

This research aims to develop a new, consistent set of axioms for Quantum Field Theory (QFT) by integrating insights from various historical approaches, thereby creating a unified axiomatic framework. This framework addresses and potentially resolves non-renormalizability and inconsistencies in perturbative methods. Additionally, the project seeks to offer a deeper and more comprehensive understanding of fundamental interactions in physics, advancing theoretical knowledge, addressing Hilbert's 6th problem and paving the way for future research and practical applications in fundamental physics.

Mathematical Techniques Required

Topics like algebraic & geometric aspects of quantum and classical theory and non-commutative geometry. A rich understanding of topology, measure theory, functional analysis, differential geometry, and spectrum theory is needed.

Motivation and My capabilities

The motivation for this research stems from an extensive background in quantum foundations, reflected in my previous undergraduate and master's research. The need for a second wave of mathematical rigour in QFT is analogous to von Neumann's formalization of quantum mechanics.

I am a highly motivated student with experience in independent research. My research interests span the Reconstruction and Foundations of Quantum Theory, Mathematical Aspects of Physical Theories, and Unification Attempts. Broadly, it lies in theoretical and mathematical physics, focusing on quantum theory. My background includes undergraduate research in Quantum Mechanics, Quantum Foundations, Relativistic theories and Newtonian Gravity. I took a Quantum Field Theory (QFT) course and advanced mathematical methods, such as Singular perturbation theory, Integral transforms, and Real & Complex analysis.

Timeline

The following (see Table below) with tentative dates for completing PhD dissertation if granted four years long option. Shaded areas indicate the active period, and '?' depicts the 'open-ended' duration.

Plan of work	Tentative Years Of Doctoral Research														
	Self-prep.			I:2024-25			II:2025-26			III:2026-27			IV:2027-28		
	-	-	-	10	02	06	10	02	06	10	02	06	-	-	-
General Reading															
Specific Courses									?						
Review															
Specific Research Title				?		?									
Discussion		?													
Literture Review															
Drafting															
Presentation									?						
Manuscript Writing														?	
Presentation, Viva Voce														?	?

References

- [1] Michio Masujimo. *Path Integral Quantization and Stochastic Quantization*. Springer-Verlag, 2nd edition, 2009.
- [2] R. Haag. On quantum field theories. *Kong. Dan. Vid. Sel. Mat. Fys. Med.*, 29N12:1–37, 1955.
- [3] Arthur Jaffe and Edward Witten. Quantum yang-mills theory, 2000.
- [4] Detlev Buchholz. Current trends in axiomatic quantum field theory, 1998.
- [5] Christopher J. Fewster and Kasia Rejzner. Algebraic Quantum Field Theory – an introduction, 4 2019.
- [6] Fred Kronz and Tracy Lupher. Quantum Theory and Mathematical Rigor. In Edward N. Zalta, editor, *The Stanford Encyclopedia of Philosophy*. MRL, Stanford University, Winter 2021 edition, 2021.